

SASWAT K. NAYAK

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EDUCATION

Master of Science, Artificial Intelligence in 3D Photonics

September 2022-December 2024

RELEVANT COURSES

Software Engineering, Image Processing, Advanced Machine Learning, Statistics, Information Retrieval, Databases, Probability and Stochastic Processes, Regression and Time Series Analysis, Machine Learning in Materials, Algorithms and Data Structures, Linear Algebra, Graph Databases, Econometrics.

SKILLS

Javascript, Python, CI/CD, Docker, Gitlab,R, C++, SQL, Tableau, Clinical Trials, Spark, GPU, RF Regression, Angular, React, Java, Agile, KPIs, GCL, Mage, Azure, Caboodle, A/B testing, Excel, Power BI, ANOVA, Node.js, EHR Data, Google Cloud, NLP, GANs, RNN, Gen AI, Tensorflow, PyTorch, GANs, Airflow, Statistics, LLM, Pandas, Numpy, MLOps, Spacy, Large Language Models(BERT, GPT-2, T5, RoBERTa), Google TTS, GBM, XGBoost, ETL Pipelines, Flink, Langchain, Transformers, MLOps.

EXPERIENCE

Data Scientist(Full Time) Innovaccer Analytics(Digital Tech Unicorn)

July 2020-July 2022

- Convinced the clinical leaders to shift the entire knowledge base DB(PostgreSQL) containing mappings to ICD, SNOMED codes to a graph database(Neo4j) leading to 80% reduction in query time.
- Led a team of interns to extract patient data from progress notes, EHR data, achieving a 90% reduction in time and a 50% decrease in costs by implementing NLP solutions(NLTK, Spacy) and using auto-encoders for anomaly detection.
- Built statistical analysis dashboards using streamlit for hospital ratings, per capita income, readmission rates, major diseases, and number of insurance claims across different locations
- Orchestrated the automatic deployment and scaling of a containerized task queue subsystem responsible for batch processing across dozen of worker nodes- Docker, EC2.
- Used object detection methods like YOLO, SSD to quantify the quality of care provided by healthcare providers.

Projects [Visualization](#)

- Execute an end-to-end data engineering [project](#) on real-time stock market data using Kafka, Amazon Web Services (AWS), Apache Kafka, Glue, Athena, and SQL.
- Utilized LangChain and Llama2 to develop a [chatbot](#) fine-tuned on my experience. Enhanced its performance by 70% through the implementation of RAG, embeddings, vector stores, and optimized document chunking techniques.
- Engineered a high-performance file-system backed by relational databases, introducing a user-friendly web interface for comprehensive CRUD operations and authentication.
- Analyzed Lyft data and made GIS [dashboard](#) using various tools and technologies, including GCP Storage, Python, Compute Instance, Mage Data Pipeline Tool, Streamlit, BigQuery, and Looker Studio.
- Scraped Nordstrom using Selenium, rotating proxies to prevent website from scrape-blocking and multi-threading for improved scraping speeds- [Github](#).

GRA: Generative AI in 3D Photonics, Semiconductor Design

September 2022 - Present 2023 — *Madison, WI*

- Introduced novel spatial descriptors for 3D lattices that captured more information than the traditional symmetry based descriptors and outperformed them in machine learning tasks by 50%(F1 Score).
- Made a mixture of generative models capable of generating 3D lattices with a wide band gap.

Machine Learning Intern, Ushur

Nov 2020 - Dec 2020

- Designed and implemented an advanced algorithm combining Google Tesseract OCR with rule-based techniques to accurately extract customer data from insurance claim documents; securing victory in an internal hackathon.